



INTRODUCTION TO BIG DATA AND ANALYTICS

CHAPTER I

1. Introduction to Big Data

2. Classification of Digital Data, Structured and Unstructured Data

3. Classification of Analytics



“What is Big Data?”



Big Data refers to complex and large data sets that have to be processed and analyzed to uncover valuable information that can benefit businesses and organizations.

- It refers to a massive amount of data that keeps on growing exponentially with time.
- It is so voluminous that it cannot be processed or analyzed using conventional data processing techniques.
- It includes data mining, data storage, data analysis, data sharing, and data visualization.
- The term is an all-comprehensive one including data, data frameworks, along with the tools and techniques used to process and analyze the data.



Big data analytics involves examining massive, complex, and high-velocity datasets to uncover hidden patterns, correlations, and actionable insights. It utilizes specialized technologies—such as Apache Hadoop, AI, and machine learning—to analyze structured, semi-structured, and unstructured data, enabling better, data-driven decisions.

Why is Big Data Important?



1. Understand the market conditions: By analyzing big data you can get a better understanding of current market conditions. For example, by analyzing customers' purchasing behaviors, a company can find out the products that are sold the most and produce products according to this trend. By this, it can get ahead of its competitors.

Benefits of Big Data and Data Analytics



- 1. Increased Revenue and Efficiency:** Organizations can optimize operations and identify new market opportunities.
- 2. Improved Customer Experience:** Personalized recommendations (e.g., Netflix, Amazon) and better service.
- 3. Enhanced Security:** Real-time fraud detection for financial services.
- 4. Innovation:** Development of new products and services,



Types of Big Data

- a) **Structured** is one of the types of big data and By structured data, we mean data that can be processed, stored, and retrieved in a fixed format. It refers to highly organized information that can be stored and accessed from a database by simple search engine algorithms. For instance, the employee table in a company database will be structured as the employee details, their job positions, their salaries, etc., will be present in an organized manner. Conform to data schemas and data models, structured data are found in tables (columns and rows)
- b) **Unstructured** Unstructured data refers to the data that lacks any specific form or structure whatsoever. This makes it very difficult and time-consuming to process and analyze unstructured data. Email is an example of unstructured data. Structured and unstructured are two important types of big data.
- c) **Semi-structured** Semi structured is the third type of big data. Semi-structured data pertains to the data containing both the formats mentioned above, that is, structured and unstructured data. To be precise, it refers to the data that although has not been classified under a particular repository (database), yet contains vital information or tags that segregate individual elements within the data. Thus we come to the end of types of data.

Characteristics of Big Data



The 3 'V's of Big Data – Variety (formats), **Velocity** (speed of generation), **and Volume** (amount).

- a) Variety:** Variety of Big Data refers to structured, unstructured, and semi-structured data that is gathered from multiple sources. While in the past, data could only be collected from spreadsheets and databases, today data comes in an array of forms such as emails, PDFs, photos, videos, audios, SM posts, and so much more. Variety is one of the important characteristics of big data.
- b) Velocity** Velocity refers to the speed at which data is being created in real-time. In a broader prospect, it comprises the rate of change, linking of incoming data sets at varying speeds, and activity bursts.
- c) Volume** Volume is one of the characteristics of big data. We already know that Big Data indicates huge 'volumes' of data that is being generated on a daily basis from various sources like social media platforms, business processes, machines, networks, human interactions, etc. Such a large amount of data is stored in data warehouses. Thus

Business Intelligence vs Big Data



Although Big Data and Business Intelligence are two technologies used to analyze data to help companies in the decision-making process, there are differences between both of them. They differ in the way they work as much as in the type of data they analyze.

Traditional BI methodology is based on the principle of grouping all business data into a central server. Typically, this data is analyzed in offline mode, after storing Data Warehouses. The data is structured in a relational database

Business Intelligence vs Big Data



1. In a Big Data environment, information is stored on a distributed file system, rather than on a central server.
2. Big Data can analyze data in different formats, both structured and unstructured.
3. Data processed by Big Data solutions can be historical or come from real-time sources.
4. Big Data technology uses parallel mass processing (MPP) concepts, which improves the speed of analysis. With MPP many instructions are executed simultaneously, and since the various jobs are divided into several parallel execution parts, at the end the overall results are reunited and presented.

Classification of analytics



1. Descriptive analytics Descriptive analytics is a statistical method that is used to search and summarize historical data in order to identify patterns or meaning.

Data aggregation and data mining are two techniques used in descriptive analytics to discover historical data. Data is first gathered and sorted by data aggregation in order to make the datasets more manageable by analysts.

Data mining describes the next step of the analysis and involves a search of the data to identify patterns and meaning.

Analyzes past data to understand what happened

Classification of analytics



2. Predictive analytics Predictive Analytics is a statistical method that utilizes algorithms and machine learning to identify trends in data and predict future behaviors.

Uses models to forecast what could happen.

Classification of analytics



3. Prescriptive analytics Prescriptive analytics is a statistical method used to generate recommendations and make decisions based on the computational findings of algorithmic models.

Suggests specific actions to take